

Object Points Estimation in Project Management

1. Purpose of Object Points

The goal of Object Points estimation is to measure the project size in object-oriented or GUI-based software systems. It helps estimate the amount of work required to build the system.

2. Main Components of Object Points

Object Points are calculated from three main elements: Screens, Reports, and 3GL Components. Each element is assigned a weight based on its complexity.

Component	Description	Typical Weight (Points)
Screens	User interfaces the user interacts with	1–6
Reports	System-generated reports	2–8
3GL Components	Logic or function modules	3–10

3. Basic Formula

Total Object Points = Screens Points + Reports Points + Components Points

4. Example

Component	Count	Complexity	Points per Item	Total
Screens	10	Medium	4	40
Reports	5	Low	3	15
Components	3	High	8	24
			Total Object Points	79

5. Adjusted Object Points

New Object Points = Total OP × (100 - Reuse %) / 100

Example: If 30% of the system is reused → $79 \times (100 - 30)/100 = 55.3$

6. Effort Estimation

Effort (Person-Months) = New Object Points / Productivity (OP per Person-Month)

Example: $55.3 / 10 = 5.5$ Person-Months

7. Notes

- Productivity rate depends on project type and team experience.
- Models like Banker, PRINCE, and SEER–SEM add correction factors for better accuracy.
- Higher complexity in screens or reports increases points, effort, and cost.